Retinal Eye Disease Detection Using Deep Learning

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***Abstract*—Retinal fundus images are a valuable source of information for ophthalmologists to diagnose retina problems. Early detection can improve chances of cure and also prevent blindness. Retinal problems like diabetic retinopathy, retinitis pigmentosa can be diagnosed using retinal fundus images by medical experts. In recent times, machine learning research has focused on diagnosing diseases like diabetic retinopathy by extracting features and then classifying the image. In this research our goal is to automatically classify images with retinal problems from those of the healthy ones without performing any explicit segmentation or feature extraction. Rather, we use a deep learning model to automatically classify any retinal fundus image as healthy or diseased. The architecture of the network is both simple and fast. The model has been tested on two datasets, including real patient retinal fundus images obtained from a local hospital. The accuracy of this model has been found to be in the range 96.5 % to 99.7%.**

Keywords—Retinal Diagnosis, CNN, Deep Learning, Machine Learning, Automated Diagnosis

I. Introduction

Vision is one of the most important human senses, lack of which can affect productivity and independence of a person. Retinal diseases affect millions of people and may result in loss of vision if the disease is not diagnosed and treated early. Example of retinal diseases include diabetic retinopathy, age related macular disorder, glaucoma etc. Early treatment options that are available may cure or slow the onset of the disease. Patients treated get several more years of vision in their life. In India, although there are a number of hospitals and eye clinics in the cities, the doctor to patient ratio is still low. In rural areas, there is lack of both infrastructure and availability of ophthalmologists. Even community outreach programs are handicapped by the lack of trained personnel to effectively screen patients in rural areas. Remote image acquisition and diagnostic Centre's, require high cost and infrastructure. With progress in technology and image analysis, it is possible to automate the process of disease detection and refer the patient to the doctor for further consultation. A number of such clinical decision support systems have been developed especially to diagnose diabetic retinopathy, age related macular disorder using advances in digital image processing and machine learning. Although, the performance of many of these systems are comparable to human experts, they are specialized to diagnose a specific retinal disease. Many of these models identify, extract and analyze disease specific features from the retinal image. This involves good understanding of the disease and effort in engineering the features for the classifiers. In this project, we propose to develop a general classifier model that can distinguish healthy retinal images from diseased images. The proposed framework is deep learning that can automatically detect features at different levels from the training dataset of retinal images. Such a general model can be useful as a first-level screening tool especially in rural settings. This would enable early detection of retinal diseases and people who require no further consultation can avoid expensive travel and testing. This benefits both the medical community and the people in rural areas. With an enhanced user interface, the diagnostic tool can be operated by semi-skilled technicians addressing the problem of shortage of skilled medical professionals in rural areas. The retinal images are captured by a fundus camera. Although fundus cameras are still expensive, low cost cameras are being developed which are now affordable. This is a one-time set-up cost which can benefit society at large.

A. Machine Learning

"Machine learning (ML) is a method of data analysis that automates analytical model building" according to SAS Insights[12]. Machine Learning algorithms learn from data to find hidden patterns without being programmed where to look. ML has been rapidly used in diagnosis of diseases. These software algorithms work by finding patterns and features in images at different levels and mapping them to the known class of diseases. Supervised learning is now used for early detection and classification of eye diseases like cataract, conjunctivitis and diabetic retinopathy as can be seen in the research literature. In the section on related work, we mention several research studies where the machine prediction is comparable to that of human experts for specific eye diseases.

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B. Neural Network

Artificial Neural Networks (ANN) are mathematical models that copy the neural structure of the mammalian cerebral cortex but on much smaller scale[13]. Neural networks are neurons organized in layers. Layers of the ANN are made up of a number of fully connected 'nodes' which contain a non-linear 'activation function' that can be used to reduce error using back propagation while performing gradient descent. Patterns are recognized by the network via the 'input layer', which is followed by one or more 'hidden layers' where the pattern processing is done via a system of weighted 'connections'. The hidden layers then link to an 'output layer' which realizes patterns across retinal images in this case. Most ANNs contain some form of 'learning rule' which alters the weights of the neurons according to the input patterns that it is presented with. However a traditional neural network fails to analyze patterns in different positions. A Convolutional Neural Network (CNN) is a type of neural network that can identify structural features in an image. The CNN is able to capture the pattern at any location across the retina, by allowing a filter to slide through the entire image to perform pattern matching. The filter moves across the image using a stride which decides how much the filter must move as it matches the image pattern. CNNs are made up of processing units that have self-learning weights and biases. Each neuron receives some inputs, performs a dot product of these with weights and biases and optionally follows it with an activation function. The whole network uses a single differentiable score function, from the raw image pixels on one end to class scores at the other. The CNN architecture take advantage of the fact that the inputs are images, which allows to encode certain properties into the architecture. This makes the forward function more efficient to implement and vastly reduces the amount of parameters in the network. In particular, unlike a regular ANN, the layers of a CNN have neurons arranged in 3 dimensions: width, height, depth. A CNN is made up of layers (minimum 5). Every layer transforms 3d input volume of data to a 3d output volume with some differentiable function that may or may not have parameters. The CNN consists of the three components. The first one is the convolutional layer which consists of filters moving by a parameter called stride to identify patterns across the image. The second component is the max pool layer which performs down sampling to ignore unnecessary features and reduce computation. The third component is the fully connected normal dense layer like a traditional neural network required to output the result.

Deep CNN is different from a neural network in the sense that a neural network is fed all the pixels of an image to a single layer and then interconnected with another dense layer, this however tends to lead to an overfitting model if the abnormal pattern occurs at different locations/positions in the retina. The edges may be detected by certain neurons and the core by others. Using the principle of “locality”, they move through the entire image by a particular stride (stride can be thought of as steps) and ensure that different neurons obtain different data of localization of patterns(in this case abnormal patterns in the retina). This way the network learns more about the pattern than figuring out where it occurs in the image as opposed to an ordinary neural network.

The data gathered by the neurons in the first layer is then fed into the corresponding hidden layers, in a feed forward manner where each neuron provides essential information of different parts of the image. The convolutional layer reads in a 3D volume of the pixels of both the color and red free images and generates an output 3D volume of the features which is then fed into the pool layer. Sufficient noise has also been included in the image in the form of a black background to prevent the model to work well only on the control images. By introducing noise and not just signal we prevent overfitting. The features output by the pool layer are all fed into the dense or fully connected layer, where all neurons are interconnected to each other (as in a bipartite graph). The final dense layer uses number of units as the number of output classes and uses a function for multi-class classification.

Our research contribution is to develop a deep learning model (CNN) for detecting retinal eye diseases from the fundus images. This model is then tested on a dataset of images obtained from a local retinal hospital. Section II summarizes research work in applying various machine learning techniques in detection and classification of eye diseases. Section III provides details of our implementation methodology. Section IV summarizes the results of the experiment and Section V concludes the paper.

II. RELATED WORK

Eye diseases that are common in the elderly include cataract, iridocyclitis and corneal haze. Computer based intelligent system was developed to classify these eye disorders using three types of machine learning algorithms [1]. Features are extracted for each type of eye disease, which are then used by the classifiers viz. Artificial Neural Networks, Fuzzy-based classifier and Neuro-Fuzzy classifier. Rules are developed for the fuzzy based classifier which are disease specific. Further rules may overlap for other diseases and difficult to extract from images. The dataset was limited to 135 subjects. The classifier models show promising results with sensitivity of 95% and specificity of 100%. Gardner et.al [2] train a back propagation neural network to detect diabetic features from retinal fundus images for the diagnosis of diabetic retinopathy. The retinal features to be detected were blood vessel, exudates and hemorrhage. Preprocessing filters and network parameters are optimized to get the best sensitivity and specificity on the training data. The training dataset consisted of 147 diabetic and 32 normal fundus images Once the network is developed, it is evaluated against a test dataset that consisted of 200 diabetic and 101 normal images. The neural network achieved 88.4% sensitivity and 83.5% specificity for the detection of diabetic retinopathy when compared against a trained ophthalmologist.

Various algorithms and image processing techniques for diagnosing diabetic retinopathy were examined in this study [3]. Algorithms are grouped into five categories viz. preprocessing; localization and segmentation of the optic disk; segmentation of the retinal vasculature; localization of the

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macula and fovea; localization and segmentation of retinopathy. Major work in the field of disease detection concentrate mainly in the form of traditional image processing techniques which requires good understanding of the disease and image data. The paper discusses the algorithms, evolving techniques, relevant terms which are useful in the study of diabetic retinopathy.

A knowledge-based system has been developed in [4] for detection of five frequent eye diseases affecting Malaysian population. Initially, features are extracted from the images of each disease and treated as symptoms. Expert rules are then developed around these features. Finally a decision tree classifier is developed with forward chaining and depth-first search to perform the classification. The authors claim that this system is more comprehensive than other similar tools developed in the past. A graphical user interface had been developed so that the tool could be widely used.

A robust classification system for early detection and prevention of Glaucoma, a common cause of blindness in older adults, has been developed in [5]. Instead of using segmentation measurements, a data-driven approach is taken. In pre-processing step, disease independent variations like illumination, size are removed. Using pattern recognition, several features are extracted and combined to get glaucomatous features. The dataset consisted of 200 images and a success rate of 86% was achieved which is comparable to medical experts.

Recent advances in machine learning for diagnosis, detection of medical diseases has been reviewed by Sajda [6]. There has been progress in understanding and theoretical analysis of issues in developing algorithms and learning theory. The review covers recent advances in supervised and unsupervised linear methods and Bayesian inference in uncertainty. These methods have made significant difference in the detection and diagnosis of diseases in medical domain. The machine learning methodologies are described and specific application in biomedical domain is provided.

Jelinek et. al [7] developed an approach of diagnosing diabetic retinopathy using a visual dictionary of visual features that are extracted from training images. There is very little preprocessing of the retinal images but using SURF, points of interest and local descriptors are extracted for both normal and abnormal images. The dataset contained 672 non-DR images, 261 images with bright lesions and 246 images with red lesions. A visual dictionary is created using representative set of these descriptors. Using a step called quantization, feature vector is generated representing visual features of each image. These feature vectors are then used to train a 2-class SVM classifier with radial basis kernel to discriminate diabetic retinopathy from normal images. This cross-training methodology is robust against differences in retinal fundus color due to difference in races of patients. The authors target the indigenous population of several countries who have higher risk of diabetic retinopathy.

Diabetic related eye diseases are common cause of blindness. In order to reduce manual analysis by medical professionals, an automated system for preliminary analysis and diagnosis has been developed by Wang et al. [8] The system combines digital image processing and machine learning techniques to perform pattern recognition to classify diabetic retinal images. The presence of exudates and lesions that are common in diabetic eye disorders, is detected using an approach that combines brightness adjustment procedure with statistical classification method and local-feature-based verification strategy. The dataset consisted of 100 normal and 54 abnormal images and the system was 100% accurate in identifying retinal images with exudates. The accuracy was 70% in classifying normal images.

In this detailed study, Patton et. al [9], review concepts, techniques and applications of retinal digital analysis. Progress made in image analysis and pattern recognition have made it possible to automate the detection of retinal conditions such as diabetic retinopathy, macular degeneration due to age and other retinal disorders. It has furthered the understanding of the relationship between retinal microvasculature and systemic cardiovascular diseases. The paper begins with principles of retinal digital image analysis. Current techniques of automated segmentation of retinal landmarks like the optic disc, fovea and blood vessels are reviewed. Current progress in automated detection of diabetic retinopathy by detecting exudates, hemorrhages and macular oedema, is reviewed and the performance of a number of systems to date are compared. Also challenges and issues in quantitative measurements from retinal images are discussed. The measurement and use of arteriovenous ratio (AVR) has been explained. Finally, the authors conclude with the future use of fundal image analysis in telemedicine.

Burlina et. al [10] apply deep learning methodology to detect and classify age-related macular degeneration (AMD). Two deep convolutional networks, one specifically developed for detecting AMD and the other trained using transfer learning are evaluated. The dataset has over 130,000 fundus images collected over 12 year period of 4613 patients. The deep convolutional neural network method had accuracy that ranged between 88.4% and 91.6%, the area under the receiver operating characteristic curve was between 0.94 and 0.96, and kappa coefficient between 0.764 and 0.829. These results indicate substantial agreement with medical expert opinion on this dataset.

A diagnostic tool based on deep-learning has been developed for screening patients for common retinal diseases like age related macular degeneration and diabetic retinopathy[11]. The system uses transfer learning which trains a neural network with much less data set than what is required by conventional methods. Further, the diagnosis is human

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interpretable since it highlights the regions recognized by the neural network. The dataset consists of optical coherence tomography(OCT) images. Performance of the system is comparable to that of human experts. The authors demonstrate the general applicability of the system by diagnosing pediatric pneumonia using chest X rays.

III. METHODOLOGY

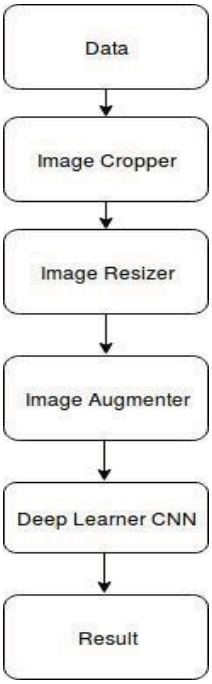

Fig. 1. Process flow

Fig 1 shows the process flow, listing the major steps involved in the implementation of the system. Each step is now described in detail.

1) Dataset: Retinal fundus images were obtained from two sources. The first set was obtained from Friedrich-Alexander University machine learning data repository [14]. The images obtained were 15 healthy images, 30 diseased retinal fundus images that included a mix of diabetic retinopathy and glaucomatous images. These images were then augmented to obtain 1021 diseased images and 522 healthy images. From this augmented source, 3 different data sets were created (a) Red free images of 128 resolution (b) Red free images of 1024 resolution (c) Color images of 1024 resolution.. Each dataset is further divided into training, validation and test dataset. The test image set consisting of 100 images was run 10 times to get average accuracy.

The second set was sourced from a local eye hospital in Bangalore, India that specializes in retinal diseases. A total of 800 diseased(Diabetic Retinopathy) and healthy images were obtained which were originally captured by a fundus camera. These images were then augmented to get 1680 images by lateral inversion and flipping. There was 960 diseased images and 720 healthy images . These had 2048 by 2048 resolution. From this augmented source 3 datasets was generated :(a) Red free images of 128 resolution (b) Red free images of 1024 resolution (c) Color images of 1024 resolution. The dataset consisted of male and female images of adults, both diseased and healthy images. Each of the above datasets was further divided into train, validation and test set using the scikit-learn library, with a random seed, to ensure similar split on multiple runs. The dataset was split as follows - 70% training and 20% validation and 10% testing.

2) Image Cropper: Images from the hospital had some patient related information that had to be masked for ensuring confidentiality and privacy. A tool was developed for the benefit of the local hospital to ensure the same. A black box was drawn over the metadata in the image, and the pixels were masked. Fig 2 shows the masking done on a retinal image.

Fig. 2. Cropped Image Without Patient Details

3) Image Resizer: The images were resized to 1024 by 1024 and 128 by 128 pixels from 2048 by 2048 for avoiding computational overhead and enable efficient training on the system. The results have been obtained for both resolutions.

4) Image Augmentation**:** The number of diseased images was more than the healthy images, Using augmentation the number of healthy images was increased.. The dataset was augmented by performing operations like rotation and flipping on existing images. The images were rotated at an angle of 10 degrees to generate new augmented images. Fig 3 shows an example of augmentation.

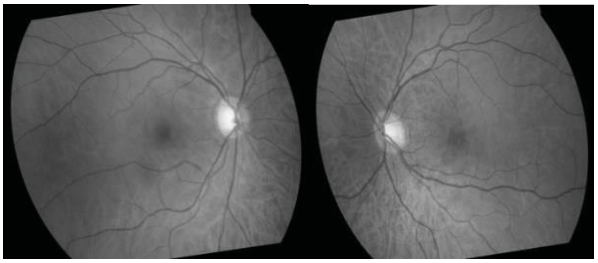

Fig. 3. Original Image (Left), Augmented Image (Right), flipped.

5) Training using Deep CNN: The images from the training dataset is first fed into the customized CNN that we refer to

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as LCDNet. LCDNet architecture is given in Fig. 4. This consists of one input layer (128x128x3), two convolutional layers(42x42x8), one max pool layer (21x21x8) and two fully connected dense layers (1x1x128), (1x1x2) supplemented by input and output layer. This model uses the power of sliding window architecture (stride hyper parameter) in a CNN to accomplish complex tasks. Eight filters are used in the convolution layers whose window size is 3x3 and stride of 1. The batch size is set to 32 and low learning rate of 0.001. The nonlinearity function used in the convolutional layer is ReLU. The pool layer uses the max function for downsampling of features to prevent computation complexity. In most runs, LCDNet performed accurately in the first epoch . In some cases, it converged in the fifth epoch.

6) Validation and hyperparameter setting: The tuning of hyper parameters is critical to the performance of the network. The weights and biases assigned by the convolutional neural network are random and can vary during different runs of the algorithm. The network was validated using the validation dataset.. On various iterations of the algorithm the batch size that was found to be ideal was 16, and the learning rate was set to a low (1/1000)value for optimum results.

7) Testing and Prediction: The test image dataset was a mix of healthy and diseased images. LCDNet trained on color, was tested on color image test set. LCDNet trained on red- free images was tested on red-free images. Each test set consisted of 100 images. There were 4 different test sets for each of the two data sources, namely 128 and 1024 pixels, both being red-free and color images of 1024 pixels. The fourth test set consisted of images from the test set which was augmented by flipping . The output/prediction of the LCDNet was written to an excel file containing the probability score of the test image belonging to one of the two classes- Normal Retina or Diseased Retina. A suitable threshold of 0.85 was chosen for this binary classification problem. The threshold was chosen based on the performance of the model averaged for 10 runs. If the model outputs 0.85 or above for either class it belongs to that particular class, a score of 0.15 or lower means it does not belong to that particular class. The excel output file now contains the label of the class and the name of the test image. Table I shows a partial list of the output file as a table.

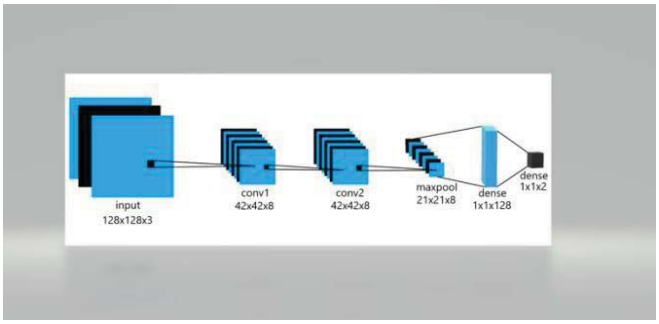

Fig. 4. LCDNet CNN Architecture

TABLE I. TABLE OF IMAGES AND THEIR PROBABILITY SCORES.

Diseased	Healthy	Test Images
5.07E-07	0.9999994928	34307_h.jpg
3.92E-05	0.9999608186	34203_h.jpg
0.9999378303	6.22E-05	48.jpg
2.94E-07	0.9999997057	34201_h.jpg
2.46E-06	0.9999975379	34305_h.jpg
0.9999381812	6.18E-05	49.jpg
0.0002147702507	0.9997852297	34411_h.jpg
6.13E-08	0.9999999387	34301_h.jpg
2.64E-07	0.999999736	34408_h.jpg
8.79E-07	0.9999912086	34205_h.jpg
0.99993564	6.44E-05	8.jpg

IV. RESULTS

The results are shown and highlighted for various types of images in table II. Augmenting (by rotation and flipping) and re-testing shows no change in results. Accuracy for red-free images is higher than color images.. Red-free images consist of only two colors hence it is easier to learn patterns across the fundus image. Accuracy for augmentation is higher than without augmentation as the dataset increases. The average accuracy over 10 runs for the dataset from the first data source is 96.52%. The average accuracy over 10 runs for the dataset from the second data source is 99.7%. The accuracy is same for both 1024 and 128 pixels resolution. Figure 5 shows results of the training and the validation phases.

```
Read train images
Train shape: (1530, 1, 24, 24) Test shape: (511, 1, 24, 24)
Read test images
iteration: 0 | Train Loss: 0.396 | Validation Loss: 0.26 | Validation accuracy(%): 91.2
iteration: 1 | Train Loss: 0.203 | Validation Loss: 0.15 | Validation accuracy(%): 94.39
iteration: 2 | Train Loss: 0.113 | Validation Loss: 0.049 | Validation accuracy(%): 98.8
iteration: 3 | Train Loss: 0.037 | Validation Loss: 0.028 | Validation accuracy(%): 99.2
iteration: 4 | Train Loss: 0.021 | Validation Loss: 0.023 | Validation accuracy(%): 99.0
```

Fig. 5. Terminal capture showing accuracy during training and validation.

TABLE II. RESULTS AS ACCURACY, PRECISION, RECALL

Type	Metric	Retinal Institute of Karnataka	Friedrich Alexander University
COLOR-1024 pixels	Precision, Recall	0.75, 0.77	0.66, 0.71
128 pixels-red-free	Precision, Recall	1.0, 1.0	0.8, 0.8
1024 pixels-red-free	Precision, Recall	1.0, 1.0	0.8, 0.8
With augmentation	Precision, Recall	1.0, 1.0	0.8, 0.8
	Average accuracy	99.70%	96.52%

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V. CONCLUSION AND FUTURE WORK

The goal of the project was to develop a system to classify the retinal fundus images as diseased or healthy. Using convolution neural networks, we have developed a system called LCDNet. that successfully performed this binary classification. Retinal fundus images was obtained from two sources, and a total of eight datasets was created for testing. We found the average accuracy of 96.5% and 99.7% on those datasets. We also found that red-free images have better results than colour, which is also the opinion of the medical community. The model can be enhanced to learn specific diseases and label the test images accordingly. This would involve getting a much larger and variety of dataset and training the model accordingly for multi-class labelling.

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